

FABRICATION NOTES:

DRC'S MUST BE RUN ON THE GERBERS PRIOR TO BUILDING BOARDS

- 1.) BOARD MATERIAL: (SEE STACKUP BELOW)

☐ RF35

☐ RO4003

☒ FR-4

☐ RO4350

☐ 4000-13EP
- 2.) BOARD STACKUP

COMPONENTS ON TOP SIDE ONLY

COMPONENTS ON BOTH SIDES

4 LAYERS

FINISHED OVERALL THICKNESS:

☐ .031 +/- .005

☒ .062 +/- .005

☐ .093 +/- .005

☐ .125 +/- .005
- 3.) COPPER WEIGHT:

EXTERNAL LAYERS: (SEE STACKUP BELOW) OUNCE FINISHED

INTERNAL LAYERS: (SEE STACKUP BELOW) OUNCE FINISHED
- 4.) PLATING:

HOT AIR SOLDER LEVEL (HASL) EXPOSED AREAS WITH LEADED SOLDER

FINISH 50uIN MIN - 200uIN MAX.

☐ FINISH: AS PER IPC ENIG, SPECIFICATION 4522, 2-4 MICRO-INCHES

IMMERSION GOLD OVER 120-240 MICRO-INCHES NICKEL

☐ OMIKRON WHITE TIN DEPOSITION PROCESS OR ANDREW APPROVED EQUIVALENT

TIN PLATING SHALL BE 200 TO 250 MICRO INCHES IN THICKNESS.

☐ STANDARD SPEC.
- 5.) DRILLING:

UNLESS OTHERWISE SPECIFIED, PLATE ALL HOLES THRU.

SEE HOLE CHART FOR FINISHED HOLE SIZES AND QUANTITY.
- 6.) HOLE SIZE TOLERANCES SHALL BE:

(SEE HOLE CHART)
- 7.) SOLDERMASK:

TOP SIDE

BOTTOM SIDE

☐ LIQUID PHOTO-IMAGEABLE SOLDERMASK SHALL BE APPLIED

OVER BARE COPPER UNLESS OTHERWISE SPECIFIED

GREEN

BLUE

RED

BLACK

☐ MATTE BLACK
- 8.) SILKSCREEN:

TOP SIDE

BOTTOM SIDE

CHARACTER LEGEND TO BE SCREENED USING NON-CONDUCTIVE

EPOXY BASED INK.

WHITE
- 9.) BOW AND TWIST: MAXIMUM ACCEPTABLE IS 0.75% (IPC-TM-650)
- 10.) FABRICATE IN ACCORDANCE WITH IPC-6011/6012, CLASS 2 (LATEST REVISIONS)
- 11.) ACCEPTABILITY TO IPC-A-600 (LATEST REVISIONS)
- 12.) CONDUCTOR INFORMATION:

DRC MUST BE RUN ON GERBERS BEFORE BUILDING BOARDS UNLESS

APPROVAL IS GIVEN IN WRITING.

SMALLEST TRACE: 6 mil

SMALLEST SPACING: 6 mil

CONDUCTOR WIDTHS AND SPACING SHALL BE WITHIN +/-20% OF ARTWORK ORIGINALS.
- 13.) REMOVE ALL BURRS AND BREAK SHARP EDGES 0.015 INCHES MAX
- 14.) VENDOR INFORMATION:

VENDOR WILL PUT THEIR UL LOGO AND DATE CODE IN SILKSCREEN

ON THE COMPONENT SIDE IN THE UNOBSTRUCTED AREA
- 15.) VIA PLUGGING/FILLING:

ALL VIAS WITH 0.2mm DRILL, 0.3mm DRILL,AND 8 MIL DRILL

SHALL BE FILLED WITH RESIN, CAPPED,

AND PLATED OVER UNLESS OTHERWISE SPECIFIED
- 16.) IMPEDANCE CONTROLS:

AS SPECIFIED

BOARD CHARACTERISTICS

Copper Layer Count:	4	Board Thickness:	1.5608 mm
Board overall dimensions:	105.9634 mm x 79.9639 mm		
Min track/spacing:	0.1524 mm / 0.1524 mm	Min hole diameter:	0.3000 mm
Copper Finish:	HAL SnPb	Impedance Control:	Yes
Castellated pads:	No	Plated Board Edge:	No
Edge card connectors:	No		

Drill Map:

- 0.200mm / 0.0079"

(25 holes)
- 0.203mm / 0.0080"

(57 holes)
- 0.300mm / 0.0118"

(89 holes)
- 0.970mm / 0.0382"

(8 holes)
- 1.000mm / 0.0394"

(5 holes)
- 1.016mm / 0.0400"

(92 holes)
- 1.100mm / 0.0433"

(4 holes)
- 1.400mm / 0.0551"

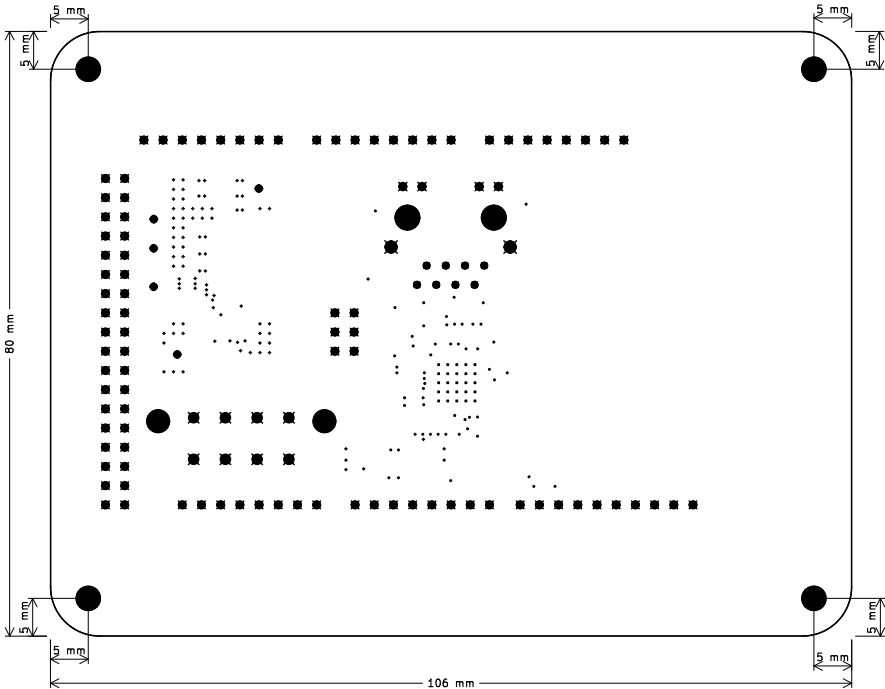
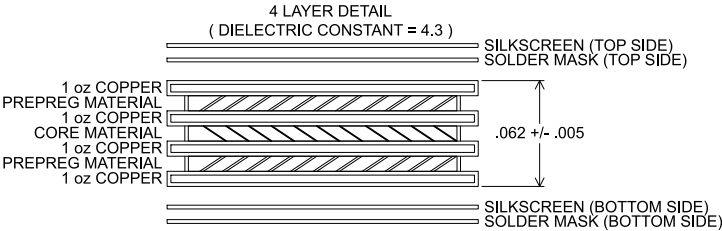
(8 holes)
- 1.650mm / 0.0650"

(2 holes)
- 3.000mm / 0.1181"

(2 holes) (not plated)
- 3.200mm / 0.1260"

(4 holes) (not plated)
- 3.260mm / 0.1283"

(2 holes) (not plated)



IMPEDANCE CONTROL REQUIREMENTS

ALL TRACES 6 MIL TRACE/6 MIL SPACE ON LAYER 4
CONTROLLED TO 100 OHMS ±10% DIFFERENTIAL, NOMINAL 50 OHM SINGLE ENDED
REFERENCE PLANE: LAYER 3 GROUND

FABRICATOR MAY ADJUST TRACE WIDTH AND DIFFERENTIAL–PAIR
SPACING AS REQUIRED TO ACHIEVE THE SPECIFIED IMPEDANCE
USING THE FINAL APPROVED LAYER STACKUP.

NEXT ASSY:		BREAK EDGES & SHARP CORNERS 0.005 MIN. TO 0.030 MAX							
PROJECT:		THIRD ANGLE PROJECTION	SCALE: SEE DWG VIEW	WBS#					

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES DIMENSIONS IN BRACKETS [XX.XX] (WHERE PRESENT) ARE MILLIMETERS AND ARE FOR REFERENCE ONLY INTERPRET DRAWING AS PER ANSI Y14.5-1994 OR Y32.2-1995		PROJECT / END ITEM: ISOLATED MULTI-RAIL POWER SUPPLY MONITOR		BOARD / ASSEMBLY: GIGA MONITOR MAIN PCB CGS-PSM-001		 Capotosto GS, Inc.			
DIMENSIONAL TOLERANCES X. = +/- 0.060 .X = +/- 0.030 .XX = +/- 0.015 .XXX = +/- 0.005	ANGULAR TOLERANCE ±0.5°	FINISH 125✓	CHECKED BY	M. CAPOTOSTO	08/05/2026	PCB FAB DRAWING MULTI-RAIL ISOLATED POWER SUPPLY MONITOR			
			VACUUM APPROVAL	N/A	N/A				
	ENGINEER APPROVAL		M. CAPOTOSTO	08/05/2026					
	SUPERVISOR APPROVAL		N/A	N/A					
	ES&H APPROVAL		N/A	N/A					
	B			QA APPROVAL	N/A	N/A	CGS-PSM-12		A
BREAK EDGES & SHARP CORNERS 0.005 MIN. TO 0.030 MAX		QA APPROVAL	N/A	N/A	SHEET SIZE	DRAWING/PART NUMBER:		REVISION	
	THIRD ANGLE PROJECTION	SCALE: SEE DWG VIEW	WBS#		ESH&O RISK LEVEL		SHEET 1 OF 1		

